

Elements Reference

Elements are XML-like tags embedded in document templates that control how data is rendered. This page is the complete reference for all available elements, their parameters, and format support.

For a conceptual introduction, see [Elements](#).

Common Parameters

Every element supports these base parameters:

Parameter	Type	Required Expression		Description
name	string	Yes	No	Identifies the element within the template
hidden	boolean	No	Yes	Suppresses the element when the expression evaluates to true. Default: false

Example — using hidden with an expression:

```
<mdoc:text name="discountLine" value="{!$concat('-', Invoice[0].DiscountPercent, '%')}" hidden="{!$Invoice[0].DiscountPercent == 0}" />
```

When `DiscountPercent` is `0`, the element is suppressed from the output entirely.

Output when `DiscountPercent` = 15:

-15%

Output when `DiscountPercent` = 0: element is hidden — nothing is rendered.

Format Support Matrix

Element	DOCX	XLSX	PPTX	Google Docs	Google Sheets	Google Slides
Repeater	✓	✓	✓	✓	✓	✓
Table	✓	—	✓	✓	—	✓
Paragraph	✓	—	—	✓	—	—
Image	✓	✓	✓	✓	✓	✓
Rich Text	✓	—	—	✓	—	—
Code	✓	✓	✓	✓	✓	✓
Text	✓	—	✓	✓	—	✓
Link	✓	—	✓	✓	—	✓
Page Break	✓	—	—	✓	—	—
Table Row	✓	—	—	✓	—	—
Table Column	✓	—	—	✓	—	—
Row	—	✓	—	—	—	—
Column	—	✓	—	—	—	—
Sheet	—	✓	—	—	—	—
Include	✓	✓	—	✓	—	—
Slide	—	—	✓	—	—	✓

Element Reference

Repeater (mdoc:repeater)

Iterates over a collection of records and renders the element body once for each record. Any other element can be nested inside a Repeater.

Parameter	Type	Required Expression		Description
value	string	Yes	Yes	The collection of records to iterate over
variable	string	Yes	No	The name used to reference the current record inside the Repeater body
mode	string	No	Yes	standard (default) — items rendered top to bottom; append — items rendered side by side

Example

```
<mdoc:repeater name="lineItems" value="Invoice[0].LineItems" variable="item">
  {!#item#.ProductName}
  {!#item#.Description}
</mdoc:repeater>
```

Each `LineItem` record in `Invoice[0].LineItems` produces one rendered block — the content between the opening and closing `mdoc:repeater` tags. The hash `#` symbol must be placed before and after the variable name — `#item#` — anywhere it's referenced inside the `Repeater`.

Output (assuming 2 line items):

Consulting Services
Website design and implementation

Support Package
Monthly technical support

XLSX-specific parameters

On Excel templates, the `Repeater` can run vertically (down rows) or horizontally (across columns):

Parameter	Type	Required	Default	Expression	Description
direction	string	No	horizontal	No	vertical — repeats rows, starting at the opening tag's column; horizontal — repeats columns, starting at the opening tag's row
lastCell	string	No	—	Yes	The end boundary for the repeat — a column letter for a vertical <code>Repeater</code> , or a row number for a horizontal one. Defaults to the last column/row in the sheet if omitted

Table (mdoc:table)

Inserts a table that iterates over a collection, repeating a defined row for each record.

Parameter	Type	Required	Expression	Description
value	string	Yes	Yes	The collection of records to iterate over
variable	string	Yes	No	The name used to reference the current record inside the table
repeatingRow	integer	No	Yes	The index of the row to repeat. Default: 1

Example

```
<mdoc:table name="invoiceLines" value="Invoice[0].LineItems" variable="line" repeatingRow="1">
  | Product | Unit Price | Quantity | Total Price |
  | {!#line#.ProductName} | {!#line#.UnitPrice} | {!#line#.Quantity} | {!#line#.TotalPrice} |
</mdoc:table>
```

The hash `#` symbol must be placed before and after the variable name — `#line#` — anywhere it's referenced inside the `Table`.

Output (assuming 2 line items):

Product	Unit Price	Quantity	Total Price	
-----	-----	-----	-----	
Consulting Services	4,800.00	1	4,800.00	
Support Package	250.00	12	3,000.00	

You can use the [Index expression](#) inside a `Table` the same way you can inside a `Repeater` — it returns the index of the current record in the iteration.

Paragraph (mdoc:paragraph)

Shows or hides an entire section of the document based on a condition. Useful for optional content blocks that should only appear when certain data is present.

Example

```
<mdoc:paragraph name="lateNotice" hidden="{!$Invoice[0].Status != 'Overdue'}">
  This invoice is overdue. Please arrange payment at your earliest convenience.
</mdoc:paragraph>
```

Output when Status = "Overdue":

This invoice is overdue. Please arrange payment at your earliest convenience.

Output when Status = "Paid": element is hidden — nothing is rendered.

On DOCX, if the Paragraph being hidden is the last one in the template, make sure it ends with an empty row — otherwise generation can throw an error.

Image (mdoc:image)

Inserts an image from a URL or base64 value. Supports dynamic image sources driven by data.

Parameter	Type	Required	Expression	Description
source	string	Yes	Yes	URL or base64 value of the image, or a merge field holding either
width	number	No	Yes	Width of the rendered image in pixels
height	number	No	Yes	Height of the rendered image in pixels

Example

```
<mdoc:image name="companyLogo" source="{!Company[0].LogoUrl}" width="200" height="80" />
```

Output: A 200×80 pixel image rendered at the element's position in the document, sourced from the URL stored in Company[0].LogoUrl.

XLSX-specific parameters

On Excel templates, an image can be rendered inside a specific cell, with control over how it fits:

Parameter	Type	Required	Default	Expression	Description
insideCell	boolean	No	false	Yes	Whether the image is rendered within a cell
mode	string	No	fit	Yes	Sizing mode, available only when insideCell is true: fit — resizes to fit the cell, preserving aspect ratio; fill — stretches or compresses to fill the cell, ignoring aspect ratio; original — keeps original size, may crop

If insideCell is true and width/height are also specified, the explicit dimensions take precedence and mode is ignored.

Rich Text (mdoc:richText)

Renders HTML-formatted content as styled text. Useful when your data source contains HTML strings that should preserve bold, italic, lists, or line breaks.

Example

```
<mdoc:richText name="invoiceNotes" value="{!Invoice[0].Notes}" />
```

Output (assuming Notes = "Net 30 payment terms apply"):

Net 30 payment terms apply

Code (mdoc:code)

Renders a barcode or QR code from a string value.

Parameter	Type	Required	Expression	Description
value	string	Yes	Yes	The value to encode
type	string	Yes	No	Barcode type — QR, CODE128, EAN13, etc.
width	number	No	Yes	Width of the rendered code image in pixels
height	number	No	Yes	Height of the rendered code image in pixels

Example

```
<mdoc:code name="paymentQR" value="{!Invoice[0].PaymentUrl}" type="QR" width="120" height="120" />
```

Output: A 120×120 pixel QR code image encoding the value of Invoice[0].PaymentUrl (e.g. https://pay.example.com/inv/0042).

XLSX-specific parameters

On Excel templates, Code uses scale instead of width/height, and adds a few Excel-specific options:

Parameter	Type	Required	Default	Expression	Description
scale	integer	No	2	Yes	Scale of the rendered code. Must be an integer greater than 0

Parameter	Type	Required	Default	Expression	Description
rotate	string	No	N	Yes	Rotation of the code image: N — normal; R — clockwise; L — counter-clockwise; I — inverted
includeText	boolean	No	false	Yes	Displays text below the code for supported code types (e.g. CODE128)
insideCell	boolean	No	false	Yes	Whether the code image is rendered within a cell
mode	string	No	fit	Yes	Sizing mode, available only when insideCell is true — same behaviour as the Image element's mode parameter

Text (mdoc:text)

Shows, hides, and formats an inline span of text within a paragraph. The body of the element holds the text to be displayed — the parameters below control how it's styled.

Parameter	Type	Required	Default	Expression	Description
font	string	No	—	Yes	Font of the element's content
size	number	No	—	Yes	Font size of the element's content
color	string	No	—	Yes	Text colour
bold	boolean	No	false	Yes	Whether the content is bold
italic	boolean	No	false	Yes	Whether the content is italic
underline	boolean	No	false	Yes	Whether the content is underlined
underlineStyle	string	No	—	Yes	Underline style. Only applies when underline is true. Available values: none, single, words, double, dotted, thick, dash, dotDash, dotDotDash, wavy, dottedHeavy, dashHeavy, dotDashHeavy, dotDotDashHeavy, wavyHeavy, dashLong, wavyDouble, dashLongHeavy
underlineColor	hex string	No	—	Yes	Colour of the underline. Only applies when underline is true

font, underlineStyle, and underlineColor may not apply when the output is a PDF, or a DOCX file opened on OneDrive online, due to format and platform limitations. A DOCX file downloaded and opened in the desktop version of Word renders the full formatting correctly.

Example

```
<mdoc:text name="overdueNotice" bold="true" color="#CC0000" hidden="{!$Invoice[0].Status != 'Overdue'}">
  This invoice is overdue.
</mdoc:text>
```

Output when Status = "Overdue": the text renders in bold, red.

Output when Status = "Paid": element is hidden — nothing is rendered.

Link (mdoc:link)

Renders a hyperlink into the document. The body of the element contains the link's display text, while the URL is set via the url parameter.

Parameter	Type	Required	Expression	Description
url	string	Yes	Yes	The URL of the link, or a merge field holding the URL

Example

```
<mdoc:link name="paymentLink" url="{!$Invoice[0].PaymentUrl}">Pay Invoice {$Invoice[0].InvoiceNumber}</mdoc:link>
```

Output (assuming PaymentUrl = "https://pay.example.com/inv/0042", InvoiceNumber = "INV-0042"):

[Pay Invoice INV-0042](https://pay.example.com/inv/0042)

Page Break (mdoc:pageBreak)

Inserts a page break at the element's position. Use hidden to conditionally suppress it.

Example

```
<mdoc:pageBreak name="termsBreak" hidden="{!$isBlank(Invoice[0].TermsAndConditions)}" />
```

Output when TermsAndConditions is not blank: a page break is inserted and subsequent content begins on a new page.

Output when TermsAndConditions is blank: element is hidden — no page break is inserted.

Table Row (mdoc:tableRow)

Used inside a Table element to remove specific rows based on a condition.

Parameter	Type	Required	Expression	Description
indexes	integer	Yes	Yes	Comma-separated row indexes to remove if hidden is true. Indexing starts at zero
hidden	boolean	No	Yes	Whether the rows listed in indexes are removed. Distinct from the general hidden parameter — this controls row removal, not the element itself

Example

```
<mdoc:tableRow name="discountRows" indexes="1,3" hidden="{!$Invoice[0].Discount == 0}" />
```

Output when Discount = 10: rows 1 and 3 render normally.

Output when Discount = 0: rows 1 and 3 are removed from the table — the overall table width is unaffected.

Table Column (mdoc:tableColumn)

Behaves the same way as Table Row, applied to columns instead.

Parameter	Type	Required	Expression	Description
indexes	integer	Yes	Yes	Comma-separated column indexes to remove if hidden is true. Indexing starts at zero
hidden	boolean	No	Yes	Whether the columns listed in indexes are removed. Distinct from the general hidden parameter — this controls column removal, not the element itself

Example

```
<mdoc:tableColumn name="internalColumn" indexes="1" hidden="{!$isBlank(Invoice[0].InternalNotes)}" />
```

Output when InternalNotes is blank: column 1 is removed from the table.

Output when InternalNotes has a value: the table renders with all columns intact.

Include (mdoc:include)

Takes elements from another template — or the current template itself — and inserts them to build a new, composite document. Useful for shared sections such as footers, terms blocks, or company headers that appear across multiple templates.

DOCX-specific parameters

Parameter	Type	Required	Expression	Description
sourceType	string	Yes	No	The type of source to pull from
source	string	Yes	No	The Document Template to pull from, or a merge field holding the ID of the source document
elementName	string	Yes	No	The specific element to pull from the source template. Only available when sourceType is a Document Template

Example — inserting one element

```
<mdoc:include name="termsSection" sourceType="documentTemplate" source="Standard Terms Template" elementName="RichText" />
```

Output: the element named "RichText" from the "Standard Terms Template" Document Template is inserted at this position, as if it were authored directly in the current template.

To include the whole document rather than a single element, select **Include the Whole Document** in the source parameter instead of picking a specific Document Template — this uses the Salesforce File source type rather than elementName.

Due to Microsoft limitations on online DOCX files, a specific element can't currently be selected from the source template when working in Word for the web — a warning appears once the source is picked. Closing the warning and inserting the element anyway includes the **whole content** of the selected Document Template. If you need a specific element, add the elementName="The Name of the needed element" attribute manually into the inserted Include element's body.

XLSX-specific parameters

On Excel templates, Include can insert a specific sheet — or an entire workbook — from another uploaded file, using a different parameter set:

Parameter	Type	Required	Expression	Description
sourceType	string	Yes	No	Type of the source document — documentTemplate or salesforceFile
source	string	Yes	Yes	The Document Template to pull from, or a merge field holding the file's ID
sheet	string	No	Yes	Name of the specific sheet to insert. Omit to insert the whole workbook

Example — inserting one sheet

```
<mdoc:include name="appendixSheet" sourceType="documentTemplate" source="Appendix Workbook" sheet="Sheet1" />
```

Output: the content of "Sheet1" from the referenced workbook is inserted, replacing the content of the receiving sheet. The receiving sheet keeps its original name.

Example — inserting a whole workbook

```
<mdoc:include name="fullAppendix" sourceType="documentTemplate" source="{!Company[0].AppendixWorkbookId}" />
```

Output: every sheet from the referenced workbook is inserted, and each takes on the name of its source sheet — the receiving sheet's own name is replaced rather than kept.

A few behaviors to keep in mind when inserting sheets or workbooks:

- Content of the included sheet overwrites data already in the receiving sheet
- Inserting a single named sheet keeps the receiving sheet's own name
- Inserting a whole workbook replaces the receiving sheet's name with each inserted sheet's original name
- If an inserted sheet's name duplicates one already in the file, a (1) suffix is appended to the inserted sheet's name

Slide (mdoc:slide)

Available in PPTX and Google Slides. Repeats a slide once per record in the collection.

Example

```
<mdoc:slide name="productSlide" value="Products" variable="product">
  {#product}
  {!product.Name}
  {!product.Description}
  {product#}
</mdoc:slide>
```

Output (assuming 3 products): 3 slides are generated from the template slide — one per product — each populated with that product's Name and Description values.

Row (mdoc:row)

Hides or collapses the row in the sheet where the element is located.

Parameter	Type	Required	Default	Expression	Description
collapseOnly	boolean	No	false	Yes	Whether the row is collapsed rather than fully hidden
hidden	boolean	No	—	Yes	Whether the row is hidden. Distinct from the general hidden parameter — this controls the row's visibility, not the element itself

Example

```
<mdoc:row name="discountRow" hidden="{!$Invoice[0].Discount == 0}" />
```

Output when Discount = 0: the row is hidden in the generated spreadsheet.

Output when Discount = 10: the row renders normally.

Column (mdoc:column)

Hides or collapses the column in the sheet where the element is located. Behaves the same way as Row, applied to columns instead.

Parameter	Type	Required	Default	Expression	Description
collapseOnly	boolean	No	false	Yes	Whether the column is collapsed rather than fully hidden
hidden	boolean	No	—	Yes	Whether the column is hidden. Distinct from the general hidden parameter — this controls the column's visibility, not the element itself

Example

```
<mdoc:column name="internalNotesColumn" hidden="{!$isBlank(Invoice[0].InternalNotes)}" />
```

Output when InternalNotes is blank: the column is hidden in the generated spreadsheet.

Output when InternalNotes has a value: the column renders normally.

Sheet (mdoc:sheet)

Hides the sheet containing the element from the workbook.

Parameter	Type	Required	Expression	Description
hidden	boolean	No	Yes	Whether the sheet is hidden. Distinct from the general hidden parameter — this controls the sheet's visibility, not the element itself

Example

```
<mdoc:sheet name="workingsSheet" hidden="{!$true}" />
```

Output: the sheet containing this element is hidden once the workbook is generated.

If a sheet-hiding expression evaluates to true on the only sheet in the workbook, generation fails — a workbook must have at least one visible sheet.